
Comments on SEBI's Discussion Paper
*Strengthening of the Regulatory framework for Algorithmic Trading &
Co-location*

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1 Algorithmic trading and the role of the regulator

The SEBI discussion paper published on 5th August 2016 is titled “Strengthening of the regulatory framework for algorithmic trading and co-location”. It adds to previous discussion and proposals from the regulator has made in the past on intervening in the growth and development of algorithmic trading practices in the Indian securities markets. Any proposal for regulation must identify the market failure that the regulation seeks to address.

This is lacking in the proposed interventions of this note. For instance, Section 1 of the paper states the objective:

to explore and address concerns related to *market quality*, *market integrity*, and *fairness* due to increased usage of Algorithmic Trading & Co-location in Indian securities market.

However, the paper does not elaborate or discuss on what the concerns are, how these concerns amount to a market failure, or even how they arise due to increased usage of algorithmic trading (AT) and co-location. Further, in Section 4.1, the paper states that

... issues that have been drawn regulatory attention are contribution to price volatility, market noise (excessive order entry and cancellation), cost that *high-frequency trading* imposes on other market users, *technological arms race*, limited opportunities for regulators to intervene during high volatility, strengthening of surveillance mechanism, etc.

However, the paper does not demonstrate that these issues exist in the case of the Indian securities markets. For example, [Aggarwal and Thomas \(2014\)](#) asks whether higher AT increases the probability of flash crash in the market, which many fear to be a consequence of AT. The evidence is that **after** the introduction of co-location and penetration of AT, the incidence of flash crashes has *reduced* for the stocks with high AT. [Brogaard et al. \(2016\)](#) show similar findings in NASDAQ data.

For example, not all AT is high frequency trading (HFT). Yet the proposals in the SEBI paper seek to disincentivise liquidity providers or to slow how HFT. Similarly, there is no evidence that there is a technological arms race that is taking place in the Indian equities markets, or why it is a issue that requires regulatory attention in India.

1.1 How regulation can be important when dealing with algorithmic trading

The benefit of algorithms to assimilate data from securities markets and place orders based on a pre-defined strategy, has been a cause for much debate on how it has changed access and quality on these markets. At the same time, developments in information technology has facilitated a disruption in the pace of the use of algorithms in securities markets: the use of algorithmic trading has increased traded volumes significantly over a short period of time. Disruptions of this nature have widened the gap between those who have been forward in investing to use such technology and those who have waited to adopt the new technology fully.

When such schisms develop between two groups in the same market, there are fears of rising inequality of access, and subsequently a drop in market quality for a subset of the market, or the market as a whole. Ultimately, this leads to an increased mistrust of the market. During such times, the role of the State is to take steps to maintain trust in the market through regulations that correct clearly identified market failure ([Zingales, 2009](#)).

The phenomenon of algorithmic trading being a disruptive force in financial markets is not specific to India. In fact, the securities markets infrastructure in economies such as the U.S. and the countries of the Euro-zone are even more vulnerable to this phenomenon because of the fragmentation of the markets across securities and derivatives, exchanges and other types of trading platforms such as ECNs. Thus, when, globally, regulators have discussed evaluating the need to intervene in the use of algorithmic trading, it has been to correct outcomes that are a combination of algorithmic trading **and** the complex, fragmented market ecosystems in these jurisdictions. Some of these have been penalising action taken against specific traders and trading firms. Of the broad market interventions that have been discussed, very little has been translated into tangible action (See Appendix [A](#)).

1.2 The impact of algorithmic trading in India

India has seen a slow and steady growth in the intensity of algorithmic trading (AT). After enabling regulation (Section 3 in the SEBI white paper), AT intensity – which is the fraction of trades where AT had some role in origination – grew from below 10 percent of the trades to around 60 percent of the trades in the markets. Since then it has remained at 65 percent on average in 2016.

During this time, there has been a significant increase in traded volumes on

average, and some improvement in liquidity as measured as the impact cost of trading. But an increase in traded volumes is not unusual for a country which has seen high GDP growth over the same time period. Thus, it becomes difficult to attribute the improvement in liquidity over this period to the increase in AT, or for any single other change in market structure.

One analysis which comes close to identifying this effect is [Aggarwal and Thomas \(2014\)](#). They examine how the increase of AT in Indian equity markets affected market quality by finding pairs of similar sized stocks, one which has high AT intensity and another with low AT intensity. They find that it is difficult to find large market cap stocks because all large cap stocks tend to have uniformly high AT intensity. However, there is considerable variation in how small and medium sized stocks have attracted AT intensity after co-location.

The analysis finds that stocks which saw a big increase in AT intensity had a significant improvement in liquidity. But there was no change or improvement in liquidity of similar sized stocks that had little or no increase in AT activity. This improvement in liquidity is strong evidence that it is AT that causes greater liquidity. In fact, it suggests that higher AT intensity is another way through which the smaller stocks can obtain higher levels of liquidity, which would traditionally only be expected to be found in the largest sized stocks in a market.

Thus, like the evidence in the research about the impact of AT and HFT in the rest of the world, research about this impact in Indian markets appear to show AT as improving market quality, and little or no evidence of an adverse impact. Section 2.6 of the paper says:

“However, the academic literature also indicate that algorithmic trading may have accentuated the issues of adverse selection costs for non-algorithmic traders and increased probability of ‘flash-crashes’ vis-a-vis the situation in the pre-algo / pre-colocation era.”

There is no reference of any empirical paper in the discussion paper which indicates the presence of the specified issues. For example, our reading of the literature reveals clear empirical evidence that adverse selection costs that disincentivise liquidity providers have *reduced* with the increased use of AT in financial markets.

There is some evidence from other countries. For example, [Hendershott et al. \(2011\)](#) use data from NYSE and show that increased AT reduces adverse selection, which in turn has given the markets lower bid-ask spread. [Brogaard et al. \(2015\)](#) study the optional co-location upgrade at NASDAQ OMX Stockholm and show that liquidity improves for the overall market and for non-co-located trading entities. After the upgrade, this paper finds that effective

spread incurred by slow trading entities fell by 2.8% after the event. *However, the market structure in every country is different. SEBI must only use Indian evidence when making regulations in India.*

The above are some examples of empirical evidence that establish the impact of AT and co-location on markets. The regulator may use these findings or conduct similar studies in order to establish what are the benefits, costs and concerns of the effect of AT on markets before designing the interventions.

2 A detailed examination of the proposed interventions

The discussion paper proposes seven measures to 'allay fear and concerns related to unfair and inequitable access to the trading system'. In the following sections, we analyse the costs and benefits of each proposal in terms of two core functions of financial markets : price discovery and liquidity.

2.1 Minimum Resting Time for Orders

The proposal is to impose a minimum order resting time of 500 milliseconds on the orders received by a stock exchange. During this period, the trader will not be allowed to modify or cancel the order. It is not clear whether the rule is proposed on all segments of equity markets, or only the spot or the derivatives segment.

The paper argues that minimum resting time could eliminate "fleeting orders", or orders that appear and then disappear in short time frames. It does not clearly define how short is the short time frame, and does not mention anything with regard to the placement of fleeting orders in the book. A fleeting order involves placing a limit order **within the bid-ask spread**, and then cancelling it within seconds (Fong and Liu, 2010).

Market abuse from fleeting orders could arise when a trader, with an intention to manipulate, places an order in the opposite direction of the trade that is genuinely desired. For example, a seller might post a small *buy* order priced **above** the current bid, in hopes of convincing other buyers to match or out-bid. If this occurs, the trader can sell a higher price (Hasbrouck and Saar, 2002). This practice is commonly known as 'spoofing'. Such orders create an artificial sense of liquidity and price in the market, and mislead other traders.¹

¹The other possible form of market abuse from fleeting orders is quote stuffing. However, there already exists an orders-to-trades ratio fee for penalising such activities.

However, there could be **legitimate** reasons for existence of fleeting orders. A trader can cancel and resubmit a limit order when the market moves away from the original limit price. Or he could switch the limit order to a market order to get immediate execution (Hasbrouck and Saar, 2009).

2.1.1 Data on fleeting orders

The discussion paper does not provide any data on existence of fleeting orders. Without knowing whether the issue of fleeting orders exists or not, it is difficult to argue for or against the proposed intervention. Using data for NSE spot and single stock futures (SSF) segment, we analyse the proportion and placement of orders cancelled in less than one second. The analysis is done for the 150 derivative securities traded on NSE for November - December 2013. The table below presents the results.

Table 1 Placement of limit orders cancelled in less than one second on NSE spot and SSF

The table presents the placement of orders cancelled that were within one second of their arrival on NSE spot and SSF market for the period of November - December 2013. The analysis is done for the top 150 derivative securities. All values are presented as percentage of total unique orders.

'Q1' represents the percentages for stocks with highest market capitalisation, and 'Q4' represents the percentages for stocks with lowest market capitalisation for the 150 securities used in the analysis.

Prices in the LOB	Q1 (Highest Market Cap)	Q2	Q3	Q4 (Lowest Market Cap)
<i>Panel A: Spot</i>				
At best price	2.47	5.22	7.12	5.14
Upto top 2 or 3	5.46	7.25	6.54	3.82
Up to top 4 or 5	5.59	5.09	3.07	1.56
Beyond top 5	23.31	10.55	5.5	2.08
Total	36.83	28.11	22.23	12.6
<i>Panel B: SSF</i>				
At best price	3.66	6.18	4.96	5.7
Up to top 2 or 3	8.19	11.18	10.75	13.04
Up to top 4 or 5	9.83	10.89	12.09	11.3
Beyond top 5	48.37	32.82	30.27	15.19
Total	70.05	61.07	58.07	45.23

Each column in the table shows the percentage of orders cancelled within a

second at different depth levels in the order book. The value '2.47' in the first row indicates that for stocks with highest market capitalisation (Q1), 2.47% of new orders on the spot market were cancelled within one second while they were quoting best prices. 23.31% of the orders were cancelled when they were beyond the best five quotes in the order book. The row 'Sum' in each panel indicates the total percentage of orders cancelled in less than one second of arrival.

The key observations from the table are:

1. Spot market experiences lower percentage of cancellations in one second than the SSF market (36.83% in Q1 stocks on spot versus 70.05% on SSF).
2. Across all stock quartiles, less than 8% of the orders are cancelled within one second while they are at best prices.
3. A majority of the cancellations that occur in less than one second of arrival take place when the order is away from top five prices in the order book.

The above observations suggest that that less than 8% of the orders could be called fleeting orders. The remaining cancellations in less than one second occur when the order is far away from the touch. We cannot further analyse the intent behind the cancellations of these orders, since that requires the knowledge of trader identity. We do not have this information in our data-set.

2.1.2 Global experience

Minimum resting time has been on regulatory list of interventions since long. The European Commission's MiFID proposed minimum quote life as an order book restriction in 2010. However, the proposal was not implemented after several consultation and commissioned reviews (Brogaard, 2011). In a working paper, (Linton *et al.*, 2012) which was published as a part of the Foresight report for the UK government, the text mentions:

"A commissioned study, (EIA3) examined the effects of minimum resting times inside a simulated market. They did not recommend its adoption."

In March 2013, the Australian securities market regulator, ASIC, proposed implementation of minimum resting time for small equity trade orders. The proposal was backed by data analysis done by the regulator. However, after extensive industry consultation, and review of data, the regulator scrapped the proposal to implement minimum resting time.

The one exchange that experimented with the implementation of minimum order resting time is the NASDAQ PSX. In 2011, NASDAQ PSX implemented a 'Minimum Life' that would commit a passive order to a minimum resting

time of 100ms, during which time it could not be cancelled or amended. As a reward, NASDAQ PSX would pay an increased rebate (additional \$0.0002 rebate per share). However, NASDAQ eventually eliminated this feature in 2013.

In 2009, the FX market platform, ICAP also introduced a 250 ms minimum order life on 5 major currency contracts, and a 1500 ms on a set of precious metals contracts. The impact of these restrictions is not documented.

In August 2015, the Toronto Stock Exchange (TSX) got regulatory approval for the introduction of a similar order type: Long life orders. In contrast to NASDAQ PSX's minimum life order, traders opting for long life orders will get priority over other orders for the same price, as a reward for committing liquidity.

NASDAQ similarly introduced 'Extended Life Order' in August 2016. According to the website, "Extended Life Orders will rest on the book for a minimum fixed resting time and receive priority over other orders entered at the same price. During this time, orders could not be amended or cancelled and would be eligible for execution during the committed resting time."

In all the above cases, the imposition of the rule was not mandated by any regulator. The new order types were introduced, and made available to traders who can commit liquid and get rewarded in terms of higher priority in the order book. These experiments are examples of market solutions that are used to give an advantage to traders who may not have fast access but are willing to provide liquidity.

2.1.3 Benefits

Minimum resting time can increase the likelihood of an order for execution **if and only if** the order is placed around the best bid and ask prices. The imposition will have *no* impact on trade execution probability if such orders are placed at a depth below the top 5 levels in the order book.

2.1.4 Costs

The costs associated with the proposal are:

1. **Reduced depth:** The proposal favors informed traders over market makers. Without the ability to modify the order over a specific time interval (500 ms in this case), the adverse selection bias that a market maker suffers will increase. This increased adverse selection would disincentivise a market maker to place orders to the exchange. So, the proposal can result in reduced liquidity provision.

2. **Increased spreads:** At present, liquidity providers compete with each other to provide the best quote. This competition has benefited market participants in form of reduced spreads. With a minimum resting time, the likelihood of market participants to aggressively compete at the best bid and ask will go *down*. The effect of lower competition will be increase in spreads.
3. **Reduced liquidity around volatile periods:** During periods of high market volatility, new information enters more frequently than during normal periods. Around such periods, quoted prices can become stale in very short time intervals, based on the new information. Without the ability to modify / cancel the quotes, market participants would be reluctant to place their orders when it is **most** needed. This will make market liquidity less resilient and more unstable around high information periods due to the imposition of minimum resting time.
4. **Reduced price efficiency:** As a result of lower competition among traders to place quotes during high information periods, prices will take longer to reflect new information. This will lead to stale prices on the markets.

Besides, it is unclear if IOC orders will also be subject to the minimum resting time. If such orders continue to remain available, then there will be a movement from placement of limit orders to IOC orders by participants. This will not be an optimal solution as this will reduce the depth in the markets. If the restriction of minimum resting time is also imposed on IOC orders, these orders will lose their importance.

2.1.5 Overall assessment

The imposition of minimum resting time could lead to several unintended consequences, without offering enough benefits to compensate for the costs. It reduces the option value of limit orders, and favors aggressive traders over passive traders. This could diminish liquidity provision in the market. Besides, restrictions on modification of an order will also raise liquidity costs by increase in adverse selection and inventory risk.

Globally, no other regulator has mandated the imposition of the rule. Various market solutions have been tried in the form of 'long duration' orders which give traders priority over others for committing to liquidity provision. Such market solutions can be tried in the Indian markets. We do not recommend a blanket imposition of the rule on the entire market.

2.2 Frequent Batch Auctions

The proposal is to move from a continuous market model to a very high frequency batch auction model. Instead of matching any compatible order to

buy and sell that comes to be market instantaneously, buy and sell orders will be aggregated over a certain period, at the end of which, compatible orders would be matched and executed as trades. The stated objective behind the measure is to nullify the *latency advantage* of co-located traders.

2.2.1 Ambiguity

It is not clear whether the proposal is meant for all segments of equity markets, or any one particular segment. It is also not clear what time interval will be used for the proposed batch auctions. What information will be supplied **during** the batch auction period is also not clear. Whether only the equilibrium price will be indicated or whether the top five prices will be indicated is not clear. Further, it is also not clear if the orders will be executed on pro-rata basis, or on time priority basis.

2.2.2 Global experience

The limited experience that is available on frequent batch auctions is from Taiwan Futures Exchange (TAIFEX). In July 2002, TAIFEX moved from frequent call auctions to continuous auction. In the call auction mechanism, orders were batched for a single price every ten seconds, ruled on a volume-maximizing algorithm and price-time priority. After July 29, 2002, continuous auction began to be used for matching orders. [Cheng and Kang \(2007\)](#) analyse the impact of the new mechanism on market quality of TAIFEX using intra-day data. They find that, *“the market is more liquid, and volatility is slightly lower, under continuous auction than under call auction. Also, there is robust evidence that continuous auction improves informative efficiency.”*

2.2.3 Benefits

The proposal draws from [Budish et al. \(2015\)](#), which argues that frequent batch auctions could eliminate “sniping”, and stop the arms race to become the fastest trader on the market. The paper also claims that the measure can reduce the potential for manipulative activities such as spoofing and layering.

As mentioned in Section [2.2.1](#), there are several ambiguities in the proposed measure. Without further clarity on how the implementation of the measure is envisaged, it is difficult to assess the benefits of the proposed measure.

2.2.4 Costs

1. **Liquidity:** Since this is a proposal and has not been tested anywhere in the world, it is unclear how frequent batch auctions will affect liquidity. In the presence of one exchange offering continuous trading (example: SGX for Nifty futures) and another offering batch auctions (NSE for Nifty futures), traders will prefer immediacy and migrate to the SGX. This will exacerbate the fragmentation of a single market for the security, and can drive away liquidity from domestic markets.
2. **Cross-market arbitrage:** The implementation will require perfect time synchronicity across exchanges for cross-market arbitrage. If not, then it will raise the arbitrage risk and can adversely affect the pricing efficiency across two markets. auctions.
3. **Increased adverse selection costs:** The implementation of batch auctions could increase adverse selection costs for traders by making the quotes stale as in the case of an intervention that forces a minimum resting time. This will especially be true of high volatility periods when information arrival in the market is high ([Haas and Zoican, 2016](#)).
4. **Reduced transparency:** In a continuous market mechanism, a trader sending a market order can compute execution price for his trades. Such markets update prices in real terms and provide price discovery in its true sense. This will not be possible if markets move to frequent batch auctions model.
5. **Implementation costs:** The implementation of frequent batch auctions require incurring the cost of a structural change in the market infrastructure from continuous trading. This means that all current market institutions and processes (including the calculation of margins for clearing and risk management such as price bands and margin limits). This cost will be imposed on all market participants which makes this an expensive approach.

2.2.5 Overall assessment

The idea of frequent batch auctions to reduce market abuse and preserve market integrity has its appeal, but appears difficult to implement due to the implementation costs, and risk of liquidity withdrawal from domestic markets. Unfortunately, SEBI's stated reason for the proposed intervention is to nullify the latency advantage of co-located traders. It is not clear how much the intervention would benefit the non co-located traders.

With respect to global experiences, TAIFEX offers one case, but against the intervention. There are some experiments that are being tried by different exchanges in the form of batch auctions. These include the Chicago Stock Exchange's SNAP – CHX SNAP, which went live in May 2016, PDQ ATS on-demand auctions, the London Stock Exchange's mid-day intra-day auction of two minutes at 12:00 every day (since March 21, 2016). However, no securities

market regulator has promoted the idea of moving to frequent batch auctions probably due to the inherent risks of implementation (in terms of reduced liquidity).

The measure is bound to disrupt the current market structure. Without extensive consultation and pilot experiments, we do not recommend the introduction of frequent batch auctions.

2.3 Random speed bumps or delays in order processing / matching

As yet another intervention to nullify the latency advantage, the regulator proposes the introduction of randomized order processing delay of few milliseconds. The idea is similar to the recently approved and launched stock exchange in the US, IEX.

2.3.1 Data on average time-span between modifications

The intended objective of the proposal is to discourage latency sensitive strategies of HFT. The scope for latency sensitive strategies arises when traders use speed to gain from price discrepancies or profit opportunities that do not stay longer than a few seconds. It will be first useful to understand the latency differences between co-located and non co-located traders, and then examining whether a speed bump of some milliseconds could reduce the latency advantage.

We do not have a direct measure of latency for co-located and non co-located traders. However, we can use average time to modifications by algorithmic and non algorithmic traders to get a sense of the differences. We analyse the average time to modifications of an order by algorithmic traders and non algorithmic traders for Nifty 50 and Nifty Next set of stocks. The analysis is done for the NSE spot market for the period of 2013.

Table 2 shows the average time between modifications in an order by algorithmic traders and non-algorithmic traders. The median time between modifications of algorithmic traders is recorded as 0.21 second on Nifty stocks. The same for non algorithmic traders is 97 seconds. The table indicates a significant difference between the reaction time of algorithmic and non algorithmic traders. However, this doesn't come as a surprise since algorithmic traders are expected to be much faster than the non-algorithmic traders.

Given the difference, the question that arises is: will the speed bump of some milliseconds bring the latency experienced by algorithmic traders closer to the latency of non algorithmic traders, and will it make the market a level playing field? The answer to the question appears to be a no.

Table 2 Average time between modifications on NSE spot segment

The table presents average time between modifications for Nifty 50 and Nifty Next stocks on the NSE spot market in 2013. Algo represents average time between modifications in an order by algorithmic traders, while non-algo represents average time between modifications by non-algorithmic traders.

All values in milliseconds

	Nifty stocks		Nifty Next stocks	
	Algo	Non algo	Algo	Non algo
Q1	26	6,847	27	8,366
Median	211	97,220	371	89,943
Q3	1,400	1,638,676	2,252	1,548,837

2.3.2 Global experience

Randomized order delay has gained attention globally as a measure curb the arms race for speed. As mentioned in the discussion paper, new exchanges / existing exchanges are experimenting with this new proposed measure to attract the disadvantaged traders. Such competition propels innovation which benefits the customers (traders / investors in this case) in the form of more choices. However, no regulator around the world has mandated the imposition of randomized order delay.

2.3.3 Benefits

The mechanism of randomized speed bumps is could stop the arms race for speed. It is not clear how much it will nullify the speed advantage of HFT. *No positive impact* on any other HFT related concern is expected.

2.3.4 Costs

- 1. Uncertainty and adverse selection costs:** The randomness element will add to the uncertainty on when the order will go to the exchange, and will increase the adverse selection costs of .
- 2. Reduced / withdrawal of liquidity:** The reduced participation because of higher adverse selection costs will likely be exacerbated during high information periods and reduce liquidity to traders and investors when it is needed most. Finally, for products which are traded on international markets (such as the SGX, DGCX), such a measure is likely to drive away liquidity from domestic markets. This will lead to higher price discovery of Indian assets on foreign shores (as is happening in the case of the currency and Nifty futures).

3. **Increased price disparities across different market segments:** Since this intervention increases uncertainty for traders, and reduced participation, it would increase persistence of violation of **cross-market arbitrage** and worsen price inefficiencies across market segments.
4. **Implementation costs:** The implementation of speed bumps will a direct cost for the exchange, and will raise costs of trading to end investors.

2.3.5 Overall assessment

The benefits of the implementation of random speed bumps are very limited. The objective of reducing the latency advantage may or may not be achieved and depends on the size of delay. A very short speed bump of 1-5 milliseconds may still not be enough for a non-colocated participant. A larger speed bump of more than 100 milliseconds can risk the liquidity and increase price inefficiency on the domestic exchanges. The costs of this mechanism are too high to compensate for the limited benefit it offers.

2.4 Randomization of orders received during a period

The proposal is to reduce the latency advantage by changing the time priority of the new / modified received in a certain time period. The revised queue with new time priority will be forwarded to the order matching engine.

2.4.1 Benefits

Like the minimum resting time, such a randomization can limit the arms race for speed. This appears to be the only benefit of this proposal.

2.4.2 Costs

The costs of this mechanism are:

1. **Relevance of IOC orders:** The mechanism defeats the time priority of the present system, and will vitiate the purpose of IOC orders.
2. **Increased uncertainty:** The introduction of randomized order delay will increase the uncertainty and the adverse selection costs of the trader, especially during volatile periods.
3. **Lower liquidity:** The new constraint could deter traders from trading. This will cause liquidity to migrate to markets where such a feature does not exist, and will benefit the offshore exchanges at the cost of domestic exchanges.

4. **Reduced cross-market arbitrage:** The uncertainty regarding when an order will reach the exchange's order matching could deter the participants from executing cross-market arbitrage at short time frames. This can result in increased disparity in prices across market segments.

2.4.3 Overall assessment

The costs of randomized order delay could outweigh the benefits that might be achieved out of the implementation of this mechanism.

2.5 Maximum message-to-trade ratio requirement

The proposal requires a market participant to execute at least one trade for a set number of order messages sent to a trading venue. The objective is to increase the likelihood of a viewed quote being available for trade and reduce hyper-active order book participation. Under this mechanism, a participant will not be allowed to place a new order / modify or cancel an existing order if the limit is breached.

2.5.1 Ambiguity

The proposal also lacks clarity on the design. It is unclear at what time interval the ratio will be calculated; whether it will be measured in real time at every point of time or whether it will be accumulated over a certain time interval; whether it will be computed at the member level?

2.5.2 Empirical evidence

As the discussion paper mentions, a similar instrument has been implemented twice before on orders in the Indian equity derivatives markets. The first was by the NSE in 2009 (reduced in 2010) in order to manage excessive placement of IOC orders causing bandwidth constraints at the exchange. The second was by SEBI in 2012 (doubled in 2013) on orders that fell outside one percentage band around the LTP. [Aggarwal *et al.* \(2016\)](#) analyse the impact of these interventions and find that such interventions can be used effectively when (a) the objective is clearly stated and (b) when it is effectively designed.

It is not clear how the maximum message-to-trade ratio is better than the existing OTR fee. The main difference between the current implementation (OTR fee) and the proposed measure is in terms of the penalty. While in the existing mechanism (OTR fee), a trader has to pay a fine if she crosses the

order-to-trade ratio, in the case of the proposed measure, the trader will *not* be allowed to send orders. Even in the current implementation, the trader is not allowed to trade in the first fifteen minutes of the continuous trading session if his order to trade ratio crossed 500 on the previous day.

Besides, the discussion paper itself cites the review report commissioned by the UK Government, ([ForesightReport, 2012](#)) – which highlighted that such a mechanism may reduce depth, increase bid-ask spreads, and exacerbate liquidity withdrawal in volatile times. Given these costs, the discussion paper doesn't justify how the benefits will compensate for the costs that this measure will impose.

2.5.3 Benefits

The benefits of message to trade ratio are similar to the benefits of minimum order resting time (Section [2.1.3](#)). However, the expected benefit of message to trade ratio is likely to be lower. There could be increased likelihood of a trade in a market with a maximum order to trade ratio in comparison to the one with no such alternative mechanism. The gains from the implementation of this measure will depend on whether the maximum message to trade ratio is binding or not. Further clarity on the design of the proposed measure is required to understand whether the

2.5.4 Costs

The costs of message to trade ratio are similar to the costs of minimum order resting time (Section [2.1.4](#)) as listed:

1. **Reduced liquidity:** If the requirement to maintain the maximum message to trade ratio *becomes binding*, the imposition of this rule could constrain the traders from sending new orders or modify or cancel existing orders. The inability to send new orders could reduce liquidity provision. Such restrictions will inevitably result in stale quotes. This is likely to trigger a similar trigger of higher adverse selection costs, higher inventory risk for a liquidity provider, to disincentivise market participants to competitively place quotes at the best prices.
2. **Withdrawals during volatile periods:** The inability to send orders after breaching of the maximum limit will adversely affect the ability of traders to provide liquidity during volatile periods.
3. **Implementation costs:** The implementation would require the exchanges to monitor the message to trade ratio of each participant at a certain time interval. This will be an additional cost for the exchange, and

2.5.5 Overall assessment

The measure may be less severe than the minimum resting time rule proposed by the regulator. However, the effectiveness of this measure will depend on whether the proposed rule is a binding constraint or not. It is not clear why a separate maximum order to trade ratio is required when the Indian markets already have a similar mechanism in the form of OTR fee.

2.6 Separate queues for colo and non colo orders

The objective of this intervention is stated as being achieving a *level playing field* for co-located and non co-located traders. This is proposed by providing two separate queues for the two categories of traders such that orders are picked up from each queue separately.

2.6.1 Benefits

The discussion paper itself mentions that the benefits will be limited since the co-located participant would still be among the first to receive the market data feeds due to his proximity to the exchange, and react to an opportunity because of technological advantage. Thus, the benefits appear to be non-existent.

2.6.2 Costs

1. **Implementation costs:** The implementation of the proposed mechanism would impose **substantial** costs on the market and the economy.
2. **Two markets for two sets of investors:** The presence of two queues will result in two sets of prices at any given time for a security: one coming from co-located participants and another for non-co-located participants. Since the co-located participants will have the advantage of responding first to prices, these prices will be the most efficient prices. This will lead to liquidity migrating to only the co-located market, and killing off the market for non-co-located trading.
3. **Withdrawal of liquidity:** Such a mechanism has the potential to drive liquidity from domestic exchanges to offshore exchanges where no such constraints would apply.

2.6.3 Overall assessment

The costs of implementing this mechanism are very high with no apparent benefits. Further, no other market place in the world has presented such a proposal and leads to a recommendation to not consider for the Indian markets either.

2.7 Review of Tick-by-Tick data feed

The proposal is to provide all market participants 'Structured data' containing top 20/30/50 bid-asks at a prescribed time interval (or as real time feed) and make the market a *level playing field* for *all* market participants irrespective of their *technological* or *financial* strength.

2.7.1 Benefits

The information may be provided to all participants at the same time interval. But it is unclear as to how equality in access to data will help small traders. It is not clear who these small traders are? Are these the retail investors – who probably do not care about continuous data feed, or are these the non-algorithmic traders.

A trader can benefit from real time feeds only if he has the infrastructure to assimilate the huge data and act upon it. Inability to process this information will not result in any benefit from elimination of TBT data and provision of real time feeds to all traders.

2.7.2 Costs

If this data feed is provided as anything other than the real time feed, the move can reduce the level of transparency in the markets. Traders will not be able to see the price situation at every given point which will adversely impact the price discovery process in the market.

2.7.3 Overall assessment

The benefits of the intervention in form of free data feed at a certain time interval are not clear. There are certainly significant costs associated with the implementation of this measure.

3 Summary

In our assessment, the critical missing aspect of the SEBI note is a clear statement of objective. Clarity of the regulatory end objectives helps to generate a more positive response from all parties in moving towards an efficient resolution of perceived market abuse.

If we use the list of concerns that have been raised in the global regulatory marketplace in response to an increase in the use of algorithmic and high frequency trading in financial markets, we find that *four out of the seven* proposed measures by SEBI aim to contain the "latency advantage" of algorithmic and co-located traders, while *two out of the seven* deal with market abuse.

The potential costs of the measures to reduce the latency advantage significantly weigh down the benefits that they may yield. These measures on implementation could result in a permanent loss and migration of liquidity which the Indian exchanges have managed to achieve over a significant time period. The impact on liquidity due to reduction in the latency advantage for co-located traders is unlikely to be compensated by any improvement in liquidity from the non co-located traders.

The issue of market abuse in the form of fleeting orders has been addressed using two proposed measures: minimum resting time and maximum order to trade ratio. However both these measures have substantial costs associated with them. SEBI must conduct data analysis to identify the percentage of fleeting orders in the market and then analyse the costs and benefits of these measures.

Concerns about higher level of systematic risk as a consequence of algorithmic trading, co-located trading or high-frequency trading are not addressed in this paper. If this is not a concern, what SEBI needs to ensure is that the risk management practices at the level of exchanges and members are robust, rather than propose regulations for structural changes in the market. Instead, this can be done through systematic review of the current pre-trade and post-trade risk management controls.

Rather than measures to drive out or reduce the size of algorithmic trading, what most exchanges globally are trying to do is to incentivise what is considered to be more desirable order placement outcomes. Some examples include innovative methods such as the introduction of the long life orders at the Toronto Stock Exchange, and extended life orders at NASDAQ. Similarly, new market designs are being tried at more recently operationalised exchanges. All these are market solutions that are being used to combat the advantage that co-located traders have in the markets.

It will be useful to similarly promote competition and market innovations in the Indian markets, rather than measures that distort and disincentivise trading as suggested by SEBI in this discussion paper.

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A Global regulatory response to algorithmic trading

The increasing percentage of algorithmic and high frequency trading (AT, HFT) in the total traded activity, and the public concern related to this type of trading, have prompted the regulators to undertake measures to regulate AT. While several interventions are being contemplated, regulators have taken a *precautionary* and *consultative* approach before implementing any intervention.

Here, we present some of the proposed measures and measures adopted by regulators of the US, the UK, France, Italy, Germany and the European Commission in the EU, and Australia.

A.1 The U.S.

A.1.1 Commodity Futures Trading Corporation (CFTC)

In December 2015, the CFTC released a notice of proposed rules to enhance the regulatory regime governing automated trading. These rules included a set of risk controls, transparency measures and other safeguards to ensure smooth functioning of the markets in the presence of AT and HFT.² The CFTC invited comments on the new proposed rules until March 16, 2016.

The rules are meant for a) market participants using algorithmic trading systems (ATS), who are defined as “AT Persons” in the rule-making, (b) clearing member futures commission merchants (FCMs) with respect to their AT Person customers, and (c) Designated Contract Markets (DCMs) executing AT Person orders.

The regulations for AT persons³ and their clearing members include:

1. Registration of AT persons not otherwise registered with the commission.
2. Pre-trade and other risk control measures for orders by the AT persons and their clearing members to prevent an algorithmic trading event. These controls include, but are not limited to, maximum AT Order Message frequency and maximum execution frequency per unit time; order price parameters and maximum order size limits; order cancellation and Algorithmic Trading disconnect systems; and connectivity monitoring systems for AT Persons with Direct Electronic Access.

²The proposed measures were preceded by a Concept Release on “Risk Controls and System Safeguards for Automated Trading Environments” by the CFTC in September 2013 on which public comments were invited. The new rules incorporate some of the comments that were received by the Commission.

³The Commission proposes a definition of who all comprise as AT persons.

3. Standards for testing, real-time monitoring and supervision and compliance of the ATS.
4. Setting up standards for designation and training of AT persons.
5. Requirements that AT persons and their clearing members submit compliance reports to DCMs regarding their ATs.

The proposed rules for DCMs include:

1. New risk controls for Direct Electronic Access provided by DCMs.
2. Transparency in DCM electronic trade matching platforms
3. New risk control procedures, include requiring DCMs to implement pre-trade risk controls; requiring DCMs to provide a test environment to AT persons.
4. Implement a review program for compliance reports regarding Algorithmic Trading submitted by AT Persons and clearing member FCMs
5. Require that certain books and records be maintained by such persons, and review such books and records as necessary;
6. Require DCMs to implement self-trade prevention tools, mandate their use, and publish statistics concerning self- trading;

A.1.2 The U.S. Securities and Exchange Commission (SEC)

The SEC has undertaken a series of steps overtime to prevent instability in the market that may arise from automated trading. The most significant of these are: Market Access Rule, Regulation SCI and registration of entities. These are:

Market access rule

In November 2010, the SEC approved a new rule to require brokers and dealers to have risk controls in place before providing their customers with direct access to the market. The rule prohibits broker-dealers from providing customers access with “unfiltered” and “naked” access to an exchange or alternative trading system. Specifically, the rule requires broker-dealers to

1. Create risk controls to prevent the entry of orders that exceed appropriate pre-set credit or capital thresholds, or that appear to be erroneous.
2. Create risk controls designed to ensure compliance with all regulatory requirements applicable in connection with market access.
3. Have certain risk management controls applied automatically on a pre-trade basis before orders route to an exchange or ATS

4. Maintain risk management controls and supervisory procedures under the direct and exclusive control of the broker-dealer with market access except in limited instances.
5. Establish, document and maintain a system for regularly reviewing the effectiveness of its risk management controls and for promptly addressing any issues.

In addition, on July 11, 2012, the SEC adopted Rule 613 under Regulation NMS, requiring self regulatory organizations to submit a plan to the SEC to create, implement, and maintain a consolidated audit trail (CAT). This audit trail is intended to increase the data available to regulators investigating illegal activities such as insider trading and market manipulation.

Regulation Systems Compliance and Integrity (Reg SCI)

In November 2014, the SEC adopted Reg SCI to strengthen the technology infrastructure of the US securities market. In that, it imposed requirements on certain key market participants to reduce the occurrence of systems issues and improve resiliency when systems problems do occur.

The rules require the self-regulatory organizations, certain alternative trading systems, plan processors, and certain exempt clearing agencies to

- Have comprehensive policies and procedures in place for their technological systems.
- The rules also provide a framework for these entities to
 - take appropriate corrective action when systems issues occur;
 - provide notifications and reports to the SEC regarding systems problems and systems changes;
 - inform members and participants about systems issues;
 - conduct business continuity testing; and
 - conduct annual reviews of their automated systems.

Registration of persons responsible for algorithmic trading strategies

In April 2016, the SEC approved a proposed rule by FINRA which require registration of Securities Traders of associated persons primarily responsible for the design, development or significant modification of algorithmic trading strategies, or who are responsible for the day-to-day supervision or direction of such activities.

In a similar attempt, in Mar 2015, the SEC had sought comments on

a proposal which require broker-dealers active in off-exchange market to become members of a national securities association. The amendments was intended to enhance regulatory oversight of active proprietary trading firms, such as high frequency traders.

A.1.3 SEC Director's testimony

In the testimony before the United States Senate Committee on Banking, Housing, and Urban Affairs Subcommittee on Securities, Insurance, and Investment on Mar 3 2016, Stephen Luparello, Director, Division of trading and markets highlighted the initiatives taken by the SEC to address HFT.

The testimony highlighted that,

While rolling back the technology clock is neither feasible nor appropriate, the SEC must assess the extent to which specific elements of an algorithmic trading environment may not optimally serve the interests of investors, as well the effectiveness of its regulatory oversight of high-speed and high-volume trading.

The testimony mentioned the following initiatives:

1. Additional timestamps for consolidated data feeds and data feed usage by Self Regulatory Organisations or SROs.
2. Membership requirements for dealers trading in off-exchange venues.
3. Rules to strengthen record-keeping requirements for algorithmic trading.
4. Work on recommendations to clarify the status of active proprietary traders not registered as broker-dealers.
5. Work on recommendations to address the use of aggressive, destabilizing trading strategies in conditions when they could most seriously exacerbate price volatility.
6. In addition, the Equity Market Structure Advisory Committee has formed a market quality subcommittee to consider the impact of technology on the efficiency of the markets and systemic risk.

A.2 The U.K.

The Foresight Report on the future of computer trading in financial markets commissioned by the UK government was submitted in 2012. The report concluded that computer based trading has led to benefits to the operation of markets in terms of liquidity, transactions costs, and the efficiency of market

prices. The report said that against the background of ever greater competition between markets, it is highly desirable that any new policies or market regulation preserve these benefits.

In terms of the future action plan, the report suggested the following priorities:

- Limit the possible future market disturbances by taking policy actions based on evidence.
- Strengthen and ease the financial market surveillance
- Improve the understanding on the effect of AT in both the short and long term.

The Bank of England's Prudential Regulation Authority (PRA) announced in May 2015 that it would assess the adequacy of existing risk measurement and management practices with respect to trading algorithms, including whether controls around algorithmic trading are "fit for purpose".

In February 2016, the Financial Conduct Authority (FCA) published its final rules to improve individual accountability for those carrying out wholesale activities, such as AT and HFT, in banks, building societies, and Prudential Regulation Authority (PRA) designated investment firms. These rules require trading firms to:

1. Legally assess the fitness and propriety of people to serve in those positions applying for approval and at least annually afterwards.
2. Apply a certification process to material risk-takers and front-line staff who pose a risk of significant harm to the firm or any of its customers (eg, staff who give investment or mortgage advice or who administer benchmarks).

A.3 The E.U.

A.4 HFT transaction tax in France

In 2012, France introduced a national levy on certain financial transactions with French shares and certain derivatives as well as canceled orders in the context of 'high frequency-trading'. In particular, the tax on canceled (or modified) HFT orders (0.01% of the value of the orders) applies in case the ratio of canceled orders to all orders during one trading day exceeds 80%. The tax is payable by each financial intermediary (except if undertaken in the context of market making) established in France and using automated algorithm trading characterised by the successive sending of purchase or selling orders and the modification or cancellation of the initial trading orders, respectively, within a time period of no more than half a second.

A subsequent report by the European Commission to assess the impact of the tax on trading volumes, price levels and volatility concluded that while no significant impact was seen on the volatility and price level of the stocks, a decline of 10% in liquidity was observed.

A.5 HFT transaction tax in Italy

In 2013, Italy levied a 0.02% tax on order changes and cancellations occurring within a time frame shorter than 0.5 seconds. The tax is calculated on a daily basis and is payable where - in a single trading day, the ratio between the sum of canceled and modified orders, and the sum of entered and modified orders exceeds 60% with reference to a single financial instrument. The tax is applicable from March 2013 for equities and from July 2013 for derivatives.

A.6 The German HFT Act

In May 2013, Germany enacted the Act on the Prevention of Risks and Abuse in High-Frequency Trading (HFT Act).

1. **Licensing:** The HFT Act requires that firms engaged in HFT must be licensed. HFT is defined to include each of the following four elements:
 - Trading for one's own account, or by proprietary trading firms;
 - Trading algorithmically without human intervention;
 - Trading using low-latency infrastructures; and
 - Trading that generates a high intra-day message rate.
2. **Risk Controls:** Firms that engage in HFT-type business must fulfill requirements to ensure that markets are not distorted or interrupted.
3. **Market Manipulation:** The Act broadens the definition of market abuse.
4. **Order to Trade Ratio:** The Act requires exchanges to impose, on a product-by-product basis, an excessive system usage fee and an order-to-trade ratio limit intended to prevent unnecessary messaging.
5. **Electronic Identification of Algorithmic Trading:** The Act mandates that exchanges have to implement rules requiring all exchange members to flag all algorithmically generated orders with a unique key when sent to a German exchange so as to allow the market surveillance system to allocate all orders to the generating algo.

A.7 MiFID II regulation by the E.C.

On Apr 24, 2016, the European Commission (E.C.) adopted a delegated regulation supplementing MiFID II. The key features in the regulation regarding AT / HFT include:

1. Algorithmic trading definition: a system shall be considered as having no or limited human intervention where, for any order or quote generation process or any process to optimise order-execution, an automated system makes decisions at any of the stages of initiating, generating, routing or executing orders or quotes according to pre-determined parameters.
2. HFT definition:
 - Shall consist of the submission on average of any of the following: a) at least 2 messages per second with respect to any single financial instrument traded on a trading venue, b) at least 4 messages per second with respect to all financial instruments traded on a trading venue.
 - Messages concerning financial instruments for which there is a liquid market shall be included in the calculation.
 - Messages introduced for the purpose of dealing on own account shall be included in the calculation.
 - For the calculation of high message intra-day rate in relation to DEA providers, messages submitted by their DEA clients shall be excluded from the calculations.
 - Trading venues shall make available to the firms concerned, on request, estimates of the average of messages per second on a monthly basis two weeks after the end of each calendar month taking into account all messages submitted during the preceding 12 months.

Pursuant to the above definition, Article 48 of MIFID II requires that a regulated market must:

1. Have effective systems and risk controls to ensure that its trading systems are resilient and have sufficient capacity to deal with peak order and message volumes, to be able to ensure orderly trading under conditions of severe market stress and to be fully tested to ensure such conditions are met and to have arrangements to ensure continuity of its services if there is any failure of its trading systems;
2. Have in place effective systems, procedures and arrangements to reject orders that exceed pre-determined volume and price thresholds or are clearly erroneous;
3. Be able to temporarily halt or constrain trading if there is a significant price movement in a financial instrument on that market or a related market during a short period and, in exceptional cases, to be able to cancel, vary or correct any transaction. Regulated markets are required to ensure that the parameters for halting trading are appropriately calibrated in a way which takes into account the liquidity of different asset classes and sub-classes, the nature of the market model and types of users and is sufficient to avoid significant disruptions to the orderliness of trading;
4. Have in place effective systems, procedures and arrangements, including requiring members or participants to carry out appropriate testing of algorithms and providing environments to facilitate such testing to ensure that trading

systems cannot create or contribute to disorderly trading conditions on the market and to manage any disorderly trading conditions which do arise from such trading systems;

5. Set and apply appropriate criteria regarding the suitability of persons to whom direct electronic access may be provided. Member States shall also require that the regulated market set appropriate standards regarding risk controls and thresholds on trading through such access and is able to distinguish and if necessary to stop orders or trading by a person using direct electronic access separately from other orders or trading by the member or participant. The regulated market shall have arrangements in place to suspend or terminate the provision of direct electronic access by a member or participant to a client in the case of non-compliance; and
6. Be able to identify, by means of flagging from members or participants, orders generated by algorithmic trading, the different algorithms used for the creation of orders and the relevant persons initiating those orders. That information shall be available to competent authorities upon request.

A.8 Australia

In 2013, the Australian Securities and Investments Commission (ASIC) released its consultation paper with regard to market integrity rules in the presence of HFT. The paper proposed to make new and amended Market Integrity Rules and/or issue guidance to address: excessive messaging and market noise (in particular, small and fleeting orders and order-to-trade ratios). The proposals that were open for public consultation included:

1. Prevent small orders being canceled or amended within 500 milliseconds of being submitted to the trading platform of a lit exchange market; and
2. Establish systems, policies and procedures to prevent the cancellation or amendment of small orders within 500 milliseconds of being submitted to the trading platform of a lit exchange market.

For the purposes of the above proposal, ASIC proposed to define ‘small order’ as being less than or equal to:

- \$500 value for equities traded on the ASX and Chi-X markets;
- \$500 value for contracts for difference traded on the ASX 24 market;
- 10 futures contracts for the ASX 24 market (for all contracts with the exception of the ASX SPI 200 Index Future (ASX SPI 200 Future);
- three futures contracts for the ASX SPI 200 Future.

The proposal does not apply to derivatives such as exchange-traded- options, options on futures and warrants.

Further, ASIC proposed guidance to market participants about compliance with automated order processing obligations to consider:

- Identification of order-to-trade ratios that may indicate a dysfunctional automated order processing system;
- The responsible use of automated order processing to monitor and control the generation of large numbers of orders that may either have an adverse impact on market quality and integrity, or that constitutes a breach of a market operator's requirements in respect of order-to-trade ratios; and
- The inclusion of analysis of order-to-trade ratio performance as part of a market participant's annual review or material change review processes.

However, in June 2013, ASIC released refinements to the proposed rules on dark liquidity and HFT, following extensive consultation with industry. In its press release, ASIC said that it decided to **not go ahead with proposals to rest small orders on the market for a set time or for dark orders to meet a minimum size**.

At present, ASIC does not have any specific regulation on AT/HFT. Instead, it periodically publishes a review of HFT and dark-liquidity in the Australian markets.⁴

⁴See <http://asic.gov.au/regulatory-resources/markets/market-structure/dark-liquidity-and-high-frequency-trading/>.